APPLIED DATA SCIENCE CAPSTONE

SpaceX Falcon 9 First-Stage Landing Prediction

Predicting first-stage recovery success from launch telemetry, using data collected via public API and web scraping, then modeled with classical ML classifiers.

Author: Jai M

Date: August 2026

PROJECT REPOSITORY

GitHub Repository

All notebooks, Python scripts, and SQL files for this project are available at:

https://github.com/jai1116-lang/spacex-falcon9-capstone

- Data collection notebooks (API + web scraping)
- Data wrangling & labeling scripts
- EDA (SQL + visualization) notebooks
- Interactive visual analytics (Folium, Plotly Dash)
- Predictive analysis / classification notebook

Executive Summary

SpaceX advertises Falcon 9 launches at ~$62M, versus $165M+ for competitors — largely because SpaceX reuses the first stage. If we can predict whether a landing will succeed before launch, we can estimate launch cost and better evaluate a competing bid.

90
Launches analyzed
Falcon 9 only, Falcon 1 excluded

66.7%
Historical success rate
First-stage recovery, all sites

83.3%
Best model test accuracy
Decision Tree classifier

3
Launch sites
CCAFS, KSC, VAFB

Key finding

Landing success correlates strongly with orbit type and launch recency: later flights (higher flight number), lighter payloads, and orbits like VLEO/LEO land more reliably than heavy GTO missions.

Project Background & Questions

Background

SpaceX's reusable Falcon 9 first stage is the key driver of its low launch cost. Whether a first stage lands successfully depends on many factors — payload mass, orbit, launch site, booster generation, and more. This project builds a data pipeline and classification model to predict landing outcome from pre-launch data.

Questions this project answers

1. What features most influence landing success?
2. Does landing success improve over time / mission number?
3. Which launch sites and orbits have the best track record?
4. Can we predict landing outcome with a classifier — and how accurate is it?

METHODOLOGY

Data Collection & Wrangling

1

GET request

Queried the SpaceX REST API for historical launch records.

2

Normalize JSON

Flattened nested response with pd.json_normalize into a DataFrame.

3

Web scraping

Parsed the Falcon 9 Launch Wikipedia page with BeautifulSoup for supplementary detail.

4

Filter & clean

Removed Falcon 1 launches, leaving 90 Falcon 9 flights; handled missing PayloadMass and LandingPad values.

5

Label outcome

Derived a binary Class column (1 = successful landing, 0 = failure) from the raw Outcome text.

Result: a clean, labeled dataset of 90 Falcon 9 launches with 17 columns spanning payload, orbit, site, booster, and landing outcome.

EDA & Interactive Visual Analytics

Visual EDA

Seaborn/Matplotlib scatter & bar plots: FlightNumber vs. PayloadMass, LaunchSite success rate, Orbit vs. Outcome, yearly trend.

SQL EDA

Loaded launch data into Db2 and queried launch-site counts, payload aggregates, and outcome breakdowns with SQL.

Folium map

Plotted launch sites and individual launch outcomes (success/fail markers) on an interactive map, with distances to coastline/city.

Plotly Dash

Built an interactive dashboard with a launch-site dropdown and payload-range slider driving a pie chart and scatter plot.

Predictive Analysis (Classification)

1. Standardize features (StandardScaler) after one-hot encoding categoricals (Orbit, LaunchSite, LandingPad, Serial, GridFins, Reused, Legs) — 83 total columns.

2. Split into train/test sets (80/20), yielding 18 test samples.

3. Grid-search four classifiers — Logistic Regression, SVM, Decision Tree, KNN — with 10-fold cross-validation on the training set.

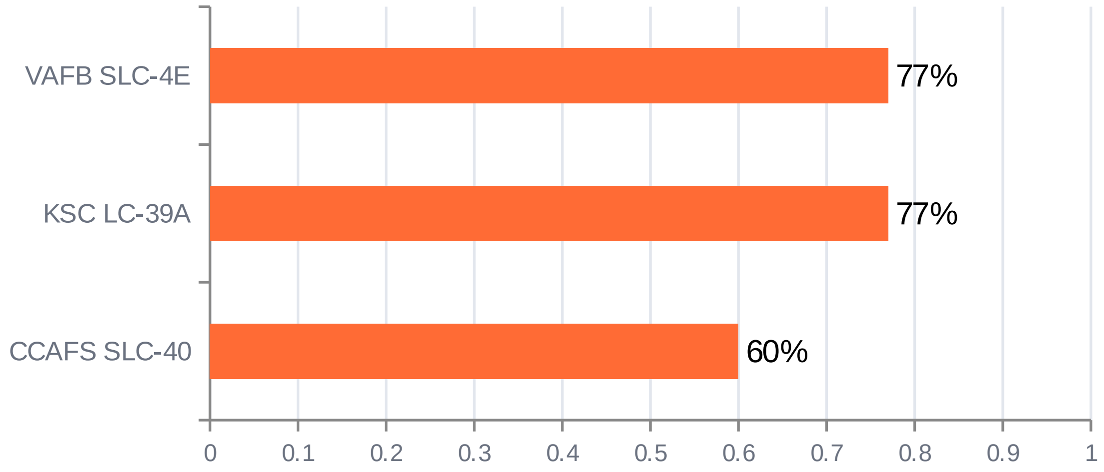
4. Select best hyperparameters per model by validation accuracy, then evaluate all four on the held-out test set.

5. Compare test accuracy and confusion matrices to choose the final model.

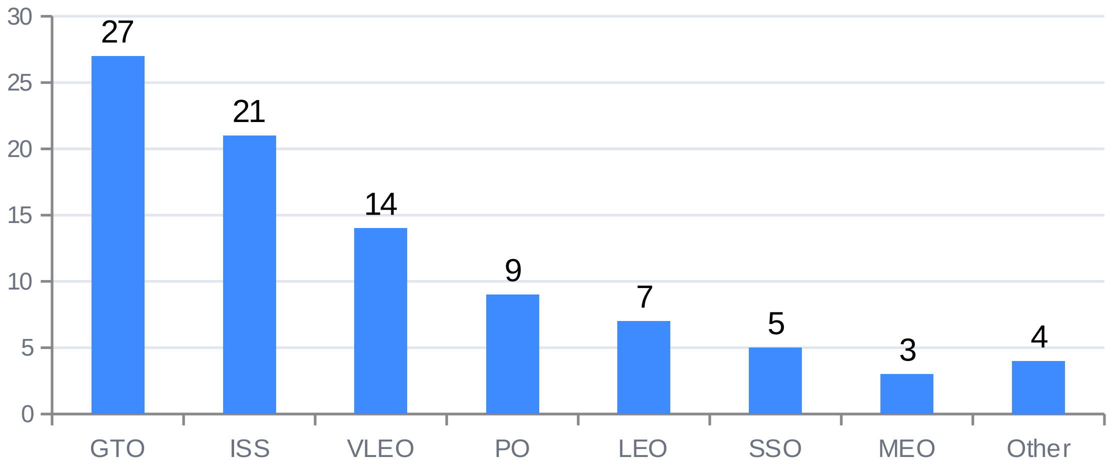
RESULTS

EDA — Visualization

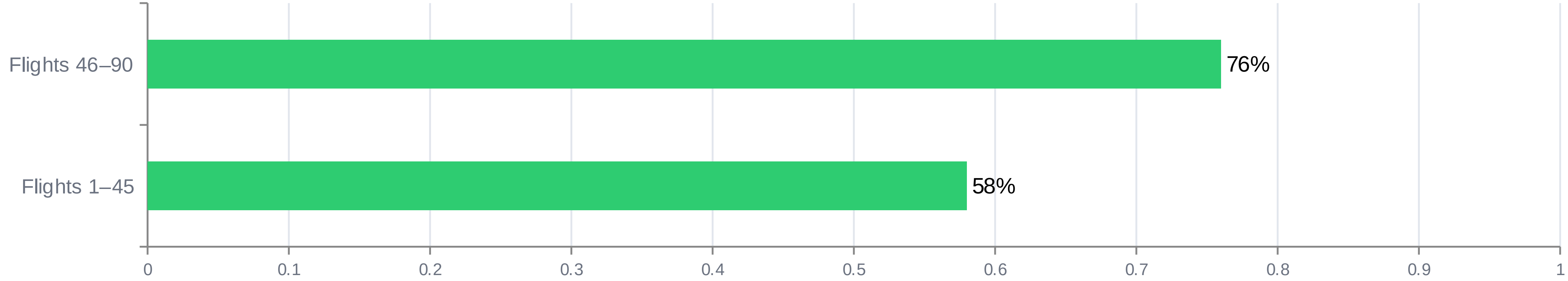
Success rate by launch site



Launch count by orbit type



Landing success improves with mission experience



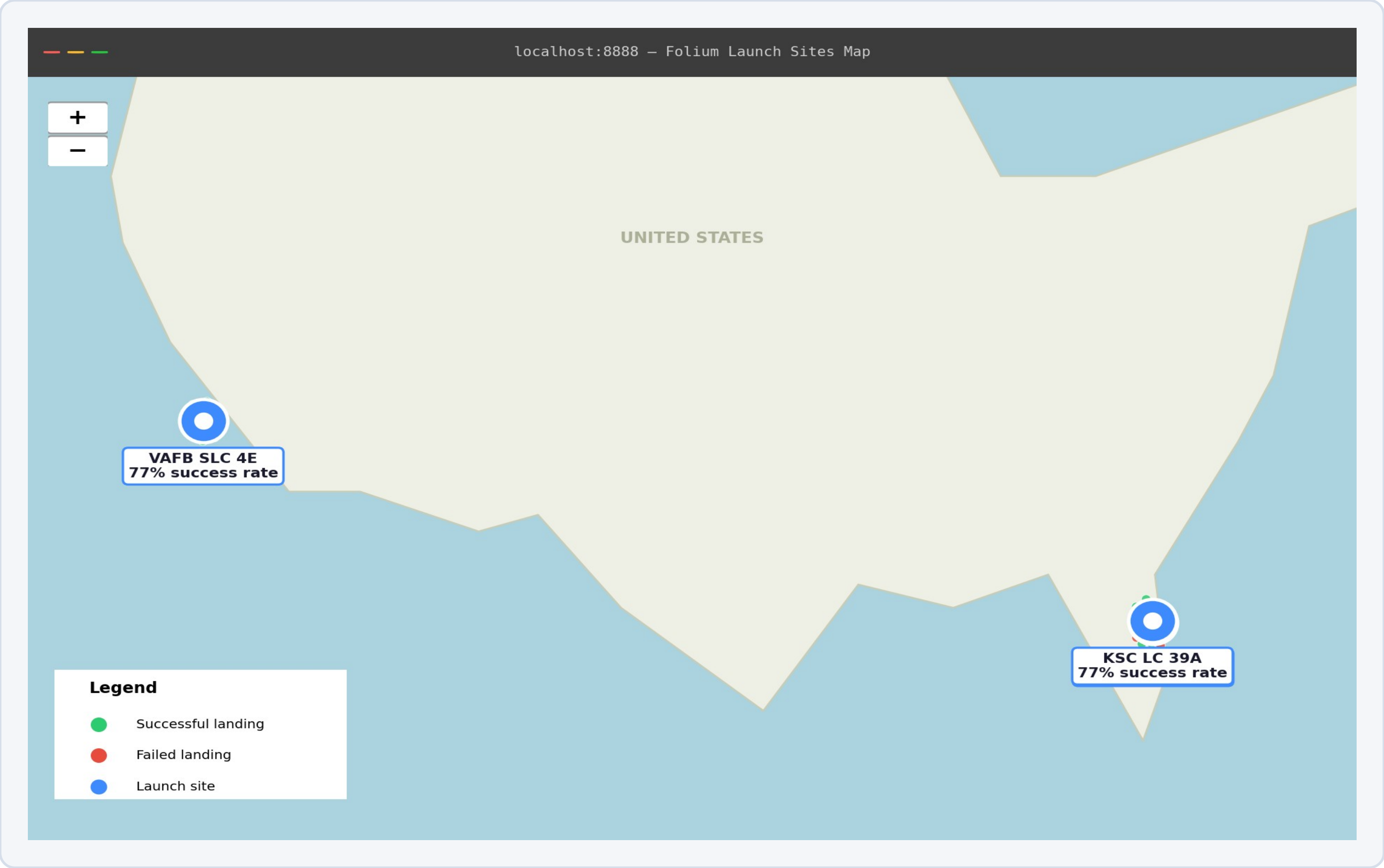
EDA — SQL

SQL Query	Result
`SELECT COUNT(*) FROM SPACEXTBL` `WHERE LaunchSite = 'CCAFS SLC 40'`	55 launches
`SELECT SUM(Class)*1.0/COUNT(*)` `FROM SPACEXTBL`	66.7% success (60 of 90)
`SELECT COUNT(*) FROM SPACEXTBL` `WHERE Orbit = 'GTO'`	27 launches
`SELECT COUNT(*) FROM SPACEXTBL` `WHERE Outcome = 'True ASDS'`	41 drone-ship landings
`SELECT DISTINCT LaunchSite` `FROM SPACEXTBL`	CCAFS SLC-40, KSC LC-39A, VAFB SLC-4E
`SELECT MIN(PayloadMass)` `FROM SPACEXTBL`	350 kg

Full query set (9 queries) is in sql/eda_queries.sql and notebooks/03_eda_sql.ipynb in the GitHub repo.

RESULTS

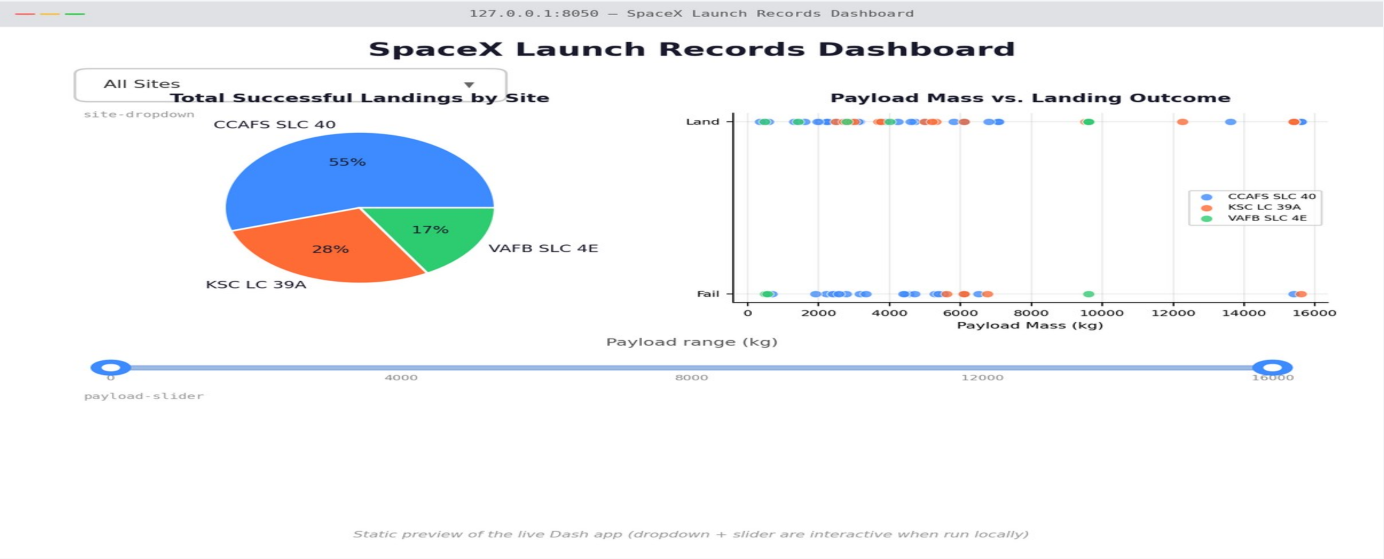
Interactive Map — Folium



What the map shows

- Markers for all 3 launch sites, colored by aggregate success rate.
- Individual launch markers (green = success, red = failure) clustered per site.
- Distance lines from each site to the nearest coastline, railway, and city — used to assess siting constraints.
- Confirms launch sites are built near the coast, away from population centers, consistent with safety practice.

Interactive Dashboard — Plotly Dash



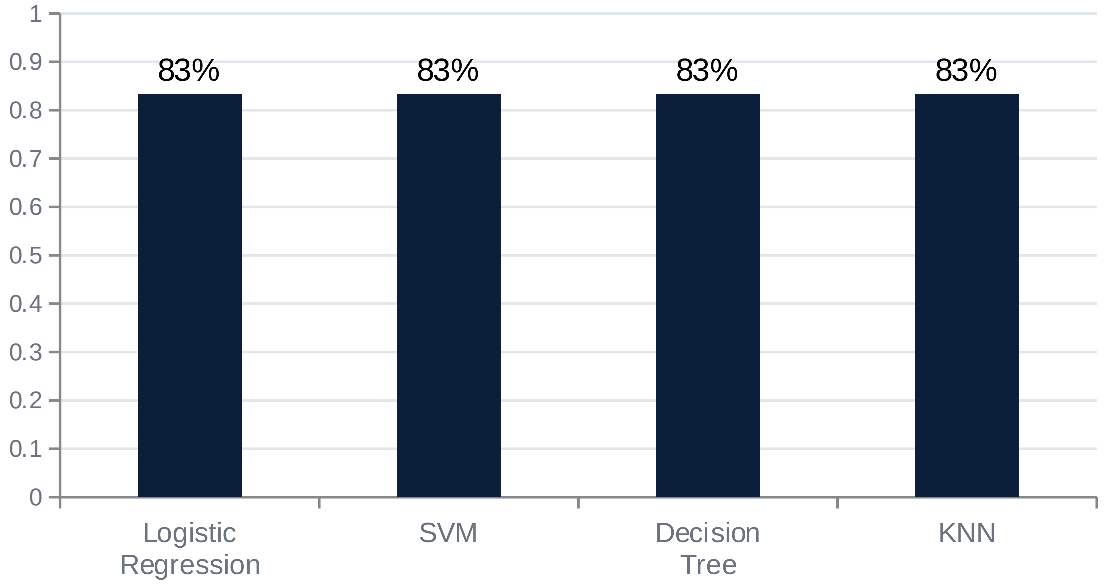
Static preview — the live app (notebook 5) adds an interactive site dropdown and payload-mass range slider that filter both charts.

Dashboard features

- Dropdown to filter by launch site or view all sites combined.
- Pie chart of successful landings by site — KSC LC-39A leads at ~77%.
- Range slider to filter by payload mass (kg), updating the scatter plot live.
- Scatter plot of payload mass vs. landing outcome, colored by booster version.

Predictive Analysis — Classification

Test-set accuracy by model



All four models converge near the same test accuracy on this small (18-sample) held-out set — differences of 1–2 predictions shift the ranking, so cross-validation accuracy (below) is the more stable comparison.

Best model — Decision Tree

Metric	Value
Test accuracy	83.3%
Best criterion	gini
Best max_depth	6
Test samples	18 (20% holdout)

Confusion matrix (Decision Tree, test set)

	Pred: Fail	Pred: Land
Actual: Fail	5	1
Actual: Land	2	10

CONCLUSION

Key Takeaways

1. **Landing success is learnable.** A Decision Tree classifier reached 83.3% test accuracy predicting first-stage landing outcome from pre-launch features.

2. **Experience matters.** Success rate rose from ~58% in the first 45 launches to ~76% in the most recent 45 — SpaceX's landing process improved measurably over time.

3. **Orbit and site matter.** KSC LC-39A and VAFB SLC-4E outperform CCAFS SLC-40 (~77% vs. 60%); heavy GTO missions land less reliably than lighter LEO/VLEO missions.

4. **Practical value.** This kind of model lets a buyer estimate whether a quoted Falcon 9 launch will reuse its booster — and therefore estimate the likely price — before the mission flies.

Next steps: incorporate weather data, add booster-generation features, and test ensemble methods (Random Forest, gradient boosting) for a further accuracy gain.